

Brief Tour of Quantum Computing

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Next three lectures...

Lecture 1

- Section 1: Turing machines, complexity classes
- Section 2: Quantum postulates *

Lecture 2

- Section 2: Quantum postulates (review)
- Section 4: Phase kickback effect

Lecture 3

- Section 4: Phase kickback effect (review)
- Section 4: Deutsch's algorithm
- Future of quantum computation

* Complex numbers, vectors, matrices, inner & outer products, tensors. Section 3: skip

The promise of quantum computing

Classical vs. quantum computability

The Turing machine is a sufficiently general model for computation to capture entirely that which can reasonably be thought of as mathematically computable.

Definition (Church-Turing Thesis CTT)

Any function that can be computed by an algorithm can be computed by a Turing Machine.

Definition (A quantum computer)

A quantum computer is a device that uses the principles of quantum mechanics to perform computation.

Are quantum computers Turing machines?

- Pauli Benioff (1980) and David Deutsch (1985) argued that any quantum computing device is still describable as a Turing machine. They established a correspondence between the time evolution of a quantum system and a Universal Turing Machine.
- These papers established an equivalence of computability between classical Turing machines and quantum computing, upholding the Church-Turing thesis

From classical to quantum Turing machines

A deterministic TM

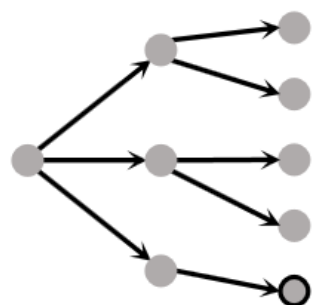


Exactly one possible action at each step

The entire history of the machine's computation is a single, linear sequence of configurations

From classical to quantum Turing machines

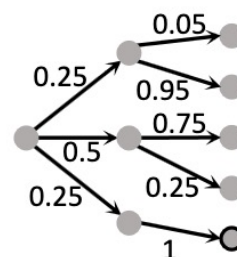
Non-deterministic TMs



A set of possible actions at each step

If the height of the tree is bounded by a polynomial in the length of the input string, then the language is in *NP*.

Probabilistic TMs

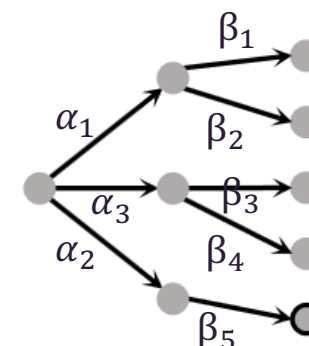


A set of possible actions at each step, with some probability

BPP is the set of languages, L , for which there is a probabilistic TM running in polynomial time with:

$$\begin{aligned} \text{Prob}(M \text{ accepts } w) &\geq 2/3, & w \in L \\ \text{Prob}(M \text{ accepts } w) &\leq 1/3, & w \notin L \end{aligned}$$

Quantum TMs



A set of possible actions at each step, associated with a complex number

BQP is the set of languages, L , for which there is a quantum TM running in polynomial time with:

$$\begin{aligned} \text{Prob}(M \text{ accepts } w) &\geq 2/3, & w \in L \\ \text{Prob}(M \text{ accepts } w) &\leq 1/3, & w \notin L \end{aligned}$$

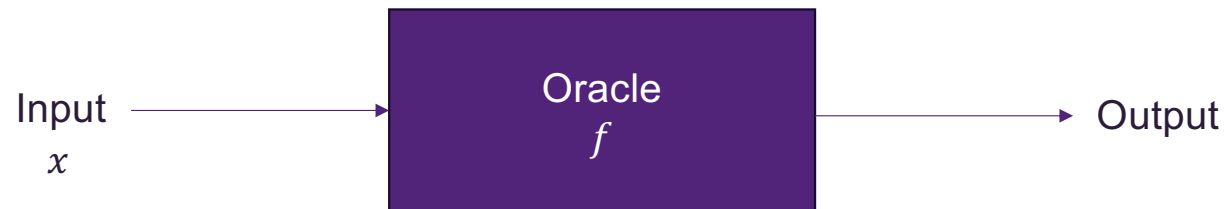
Challenging the ECTT

The ECTT:

Any algorithmic process can be simulated *efficiently* using a *Probabilistic Turing Machine*

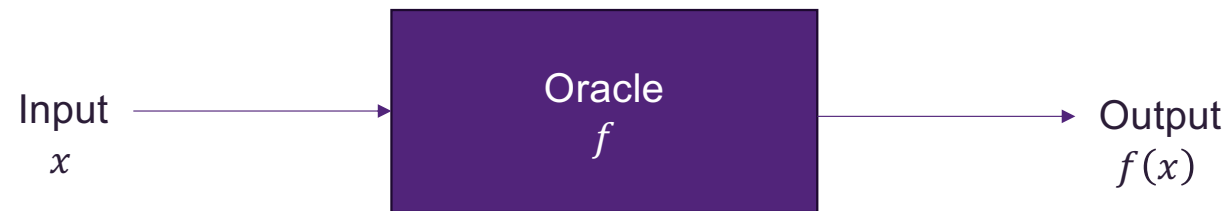
ECTT challenged by oracle problems comparing number of queries for classical vs. quantum f

Query: a single evaluation of f on input x



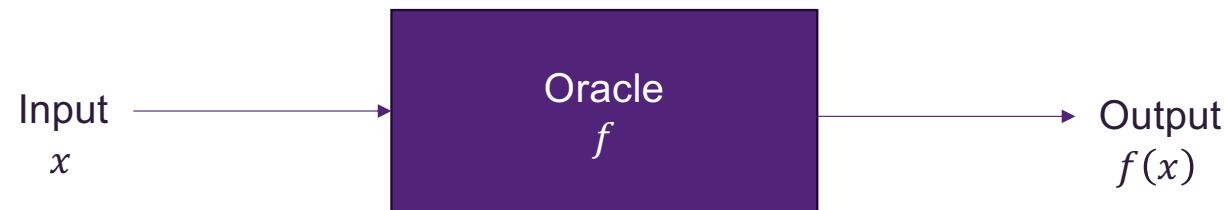
Recall: what is a promise problem?

Query: a single evaluation of f on input x



Example of a promise problem **

Query: a single evaluation of f on input x
Promise: f is constant or balanced



Example: Constant vs. Balanced functions

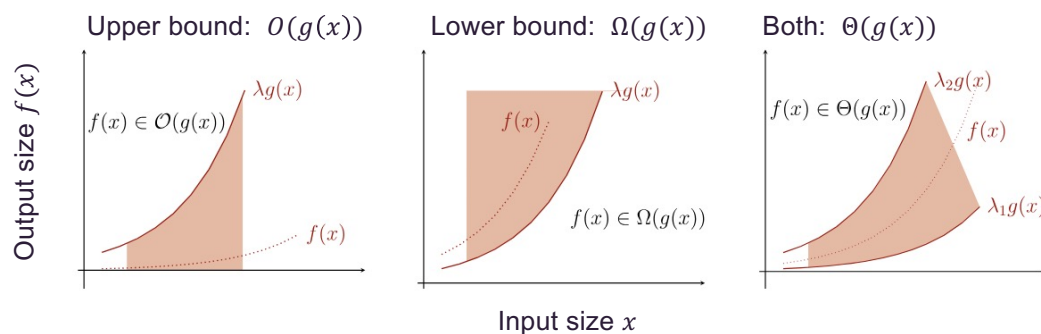
Consider a function $f: \{0,1\}^n \rightarrow \{0,1\}$ and a task whereby an algorithm must output if f is constant or balanced.

A function is *constant* if it returns the same value $f(x) \forall x$. It is *balanced* if it returns 0 for half the possible values of x , and returns 1 for the rest.

x	f_1	f_2	f_3	f_4
0	0	1	0	1
1	0	1	1	0

Challenging the ECTT

Algorithm	Speed-up	Dates	Classical	Quantum	
*Deutsch algorithm	Factor of 2	1985	2	1	x = single bit
Deutsch–Jozsa problem	Exponential	1992, 1998	$O(2^x)$	$O(1)$	x is number of bits
Factoring (Shor), DLP, period finding	Superpolynomial	1994	$O(e^{\text{polylog}(\log(x))})$	$O(\text{poly}(\log(x)))$	x is integer to be factorized
Grover search	Quadratic	1996	$O(x)$	$O(\sqrt{x})$	x is size of function domain
Bernstein-Vazirani	Linear	1997	$O(x)$	$O(1)$	x is size of function domain
Amplitude amplification and estimation	Quadratic	2000	$\Theta(x)$	$\Theta(\sqrt{x})$	x is success probability for algorithm
Quantum random walks in 1D	Quadratic	2001	$\Theta(x)$	$\Theta(\sqrt{x})$	x is mixing rate
Harrow–Hassidim–Lloyd algorithm	Exponential	2008	$O(x)$	$O(\log(x))$	x relates to variables in linear system
Simulating Coupled Classical Oscillators	Exponential	2023	$\Theta(2^x)$	$\Theta(\text{poly}(x))$	x relates to number of oscillators



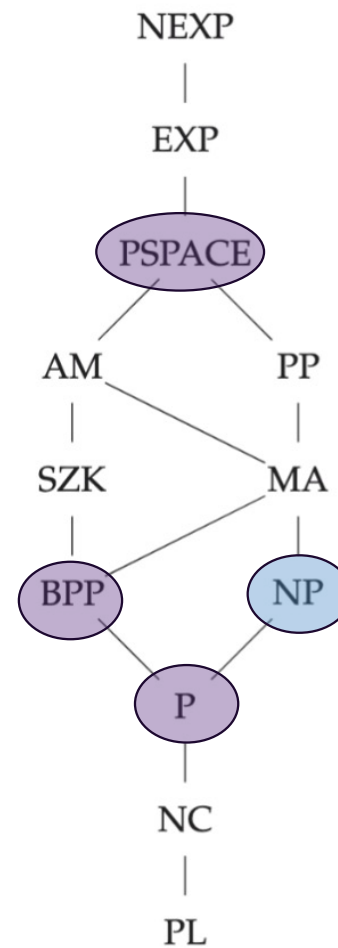
Quantity being compared changes

Classical complexity

- Each complexity class is linked by a containment $\subseteq$ going upwards.
- The class of problems that can be decided in polynomial-time on a Turing machine is denoted P
- BPP is often considered the class of problems efficiently solvable on a classical computer

$$P \subseteq BPP \subseteq PP \subseteq PSPACE$$

- Recall that we do not know if containments are strict i.e. we don't know if $P \neq NP$ or $P \neq PSPACE$



(a) Classical complexity classes.

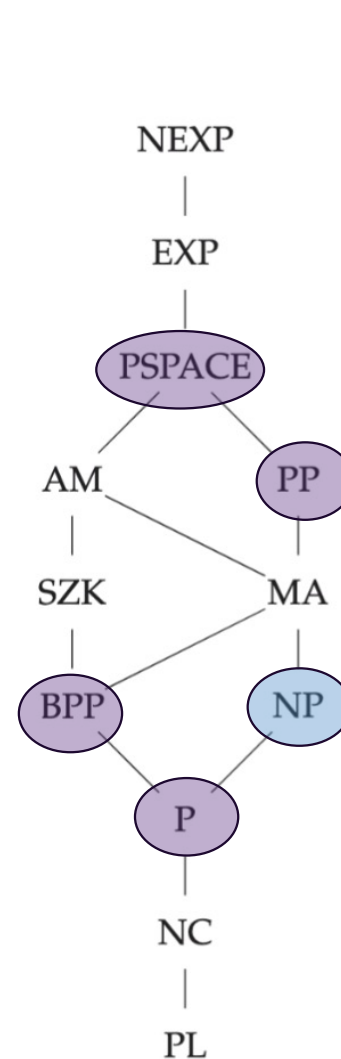
Are quantum computers provably more powerful?

- Each complexity class is linked by a containment $\subseteq$ going upwards.
- BQP (bounded error quantum polynomial time) are the class of problems efficiently solvable on a quantum computer

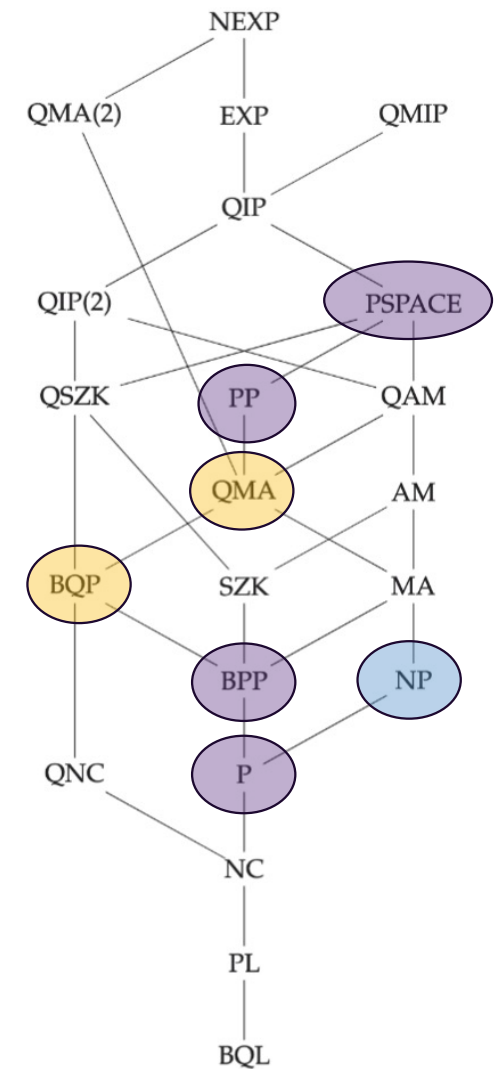
$$P \subseteq BPP \subseteq BQP \subseteq PP \subseteq PSPACE$$

- Proving that $BPP \neq BQP$ implies that $P \neq PSPACE$ which is as hard as proving $P \neq NP$.
- We don't know if containments are strict. For example, we know factoring is in BQP and that $BPP \subseteq BQP$. But we hope that factoring is not in BPP .
- We do not know how BQP sits relative to NP .

Credit: Watrous (see lecture notes for full reference list)



(a) Classical complexity classes.



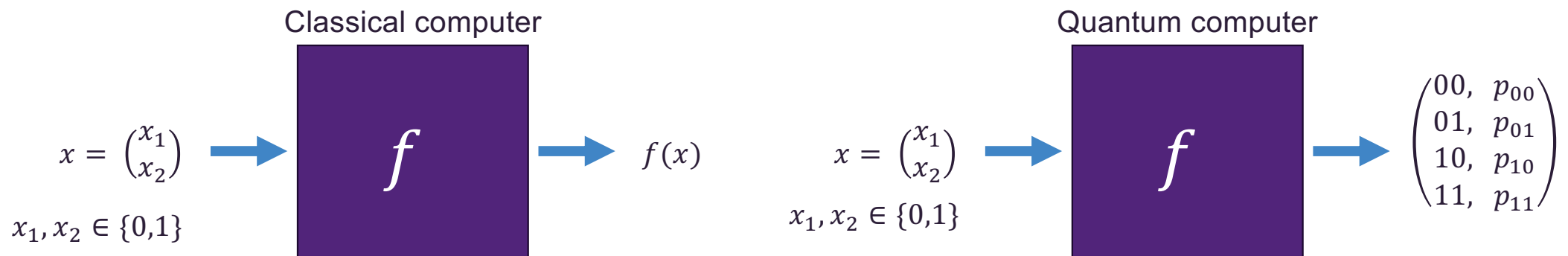
(b) Quantum and classical complexity.

Review of complexity classes

<i>Class</i>	<i>Informal definition</i>
P	Can be solved by a deterministic classical computer in polynomial time
BPP	Can be solved by a probabilistic classical computer in polynomial time
BQP	Can be solved by a quantum computer in polynomial time
NP	Solution can be checked by a deterministic classical computer in polynomial time

npj Quantum Information (2016) 2, 15023

Classical vs. quantum computing

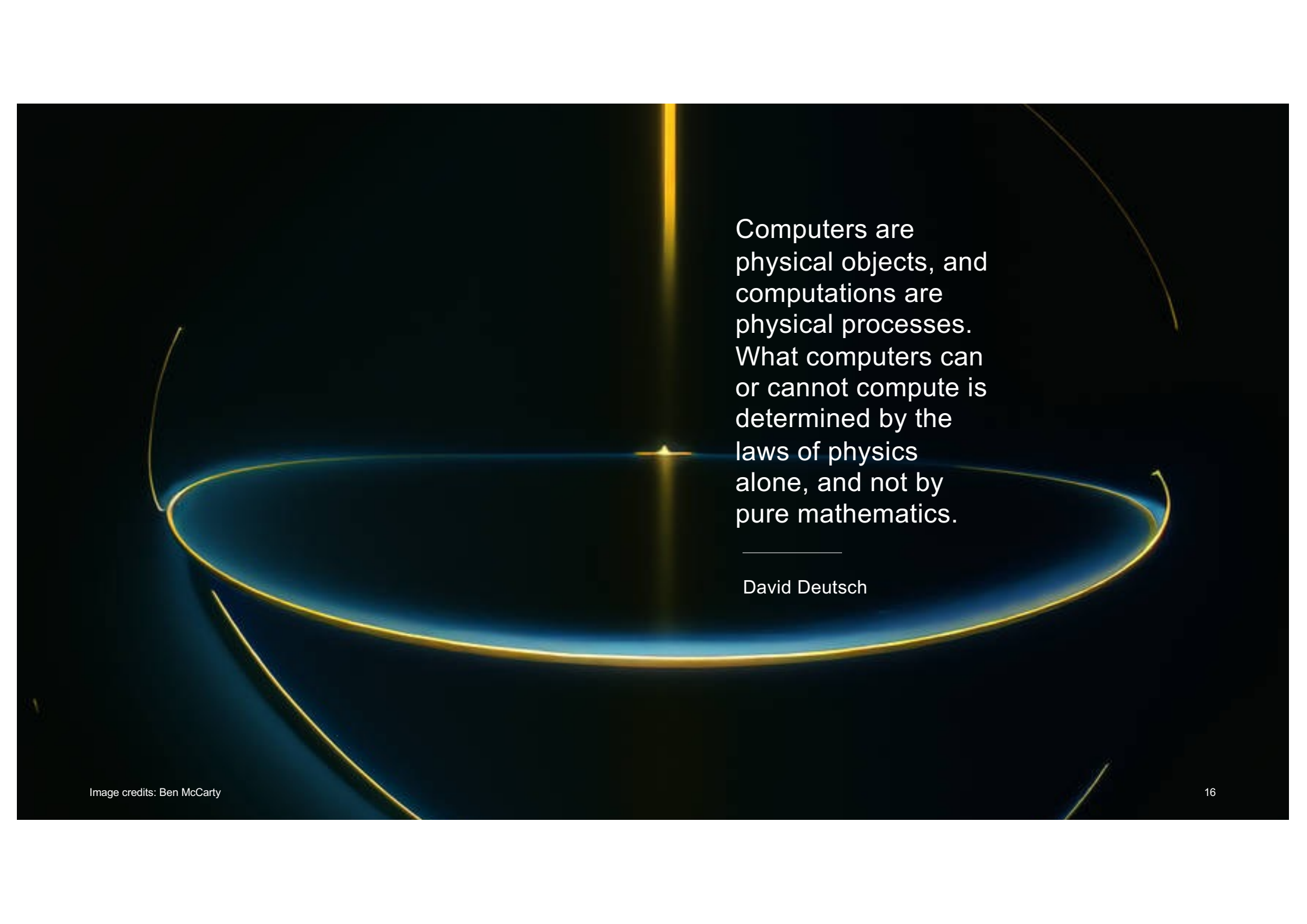


Deterministic outcome $f(x)$

Random outcomes i with prob p_i

Quantum Algorithms: Probability of finding (optimal) solutions maximized (near 1)

The physicality of computing

The background is a solid black field. A bright, vertical yellow line descends from the top center, ending at a small, glowing yellow point. From this point, a horizontal blue line extends to the left and right, forming a wide, shallow oval shape. The blue line has a yellow glow along its edges. Several thin, curved yellow lines are scattered around the central oval, some entering and exiting the frame, creating a sense of motion or orbits.

Computers are
physical objects, and
computations are
physical processes.
What computers can
or cannot compute is
determined by the
laws of physics
alone, and not by
pure mathematics.

David Deutsch

How did we make a computer from electricity?

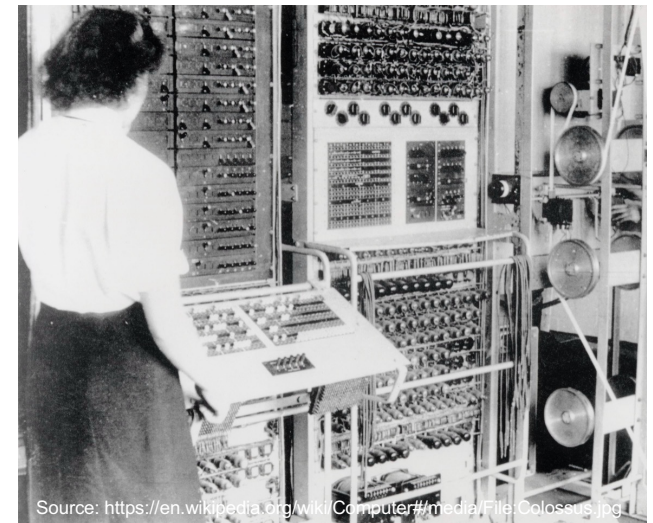
1752: Benjamin Franklin
We see things in nature



1861: Maxwell Equations
We describe things via mathematics

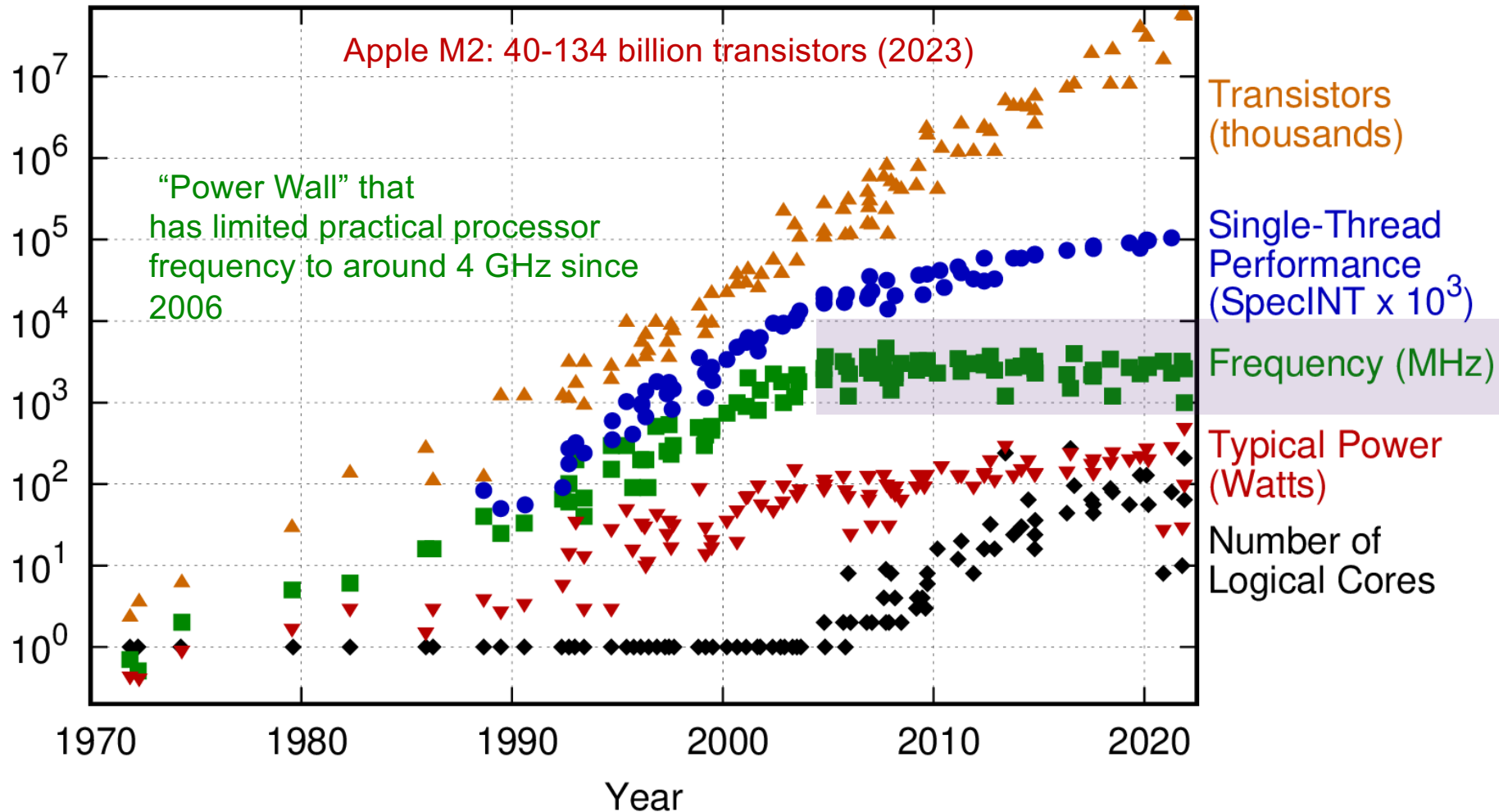
$$\begin{aligned}\nabla \cdot \mathbf{E} &= \frac{\rho}{\epsilon_0} \\ \nabla \cdot \mathbf{B} &= 0 \\ \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \nabla \times \mathbf{B} &= \mu_0 \left(\mathbf{J} + \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right)\end{aligned}$$

1943: First electronic computer
We build the hardware



It took us 200 years to control electricity to
make computers

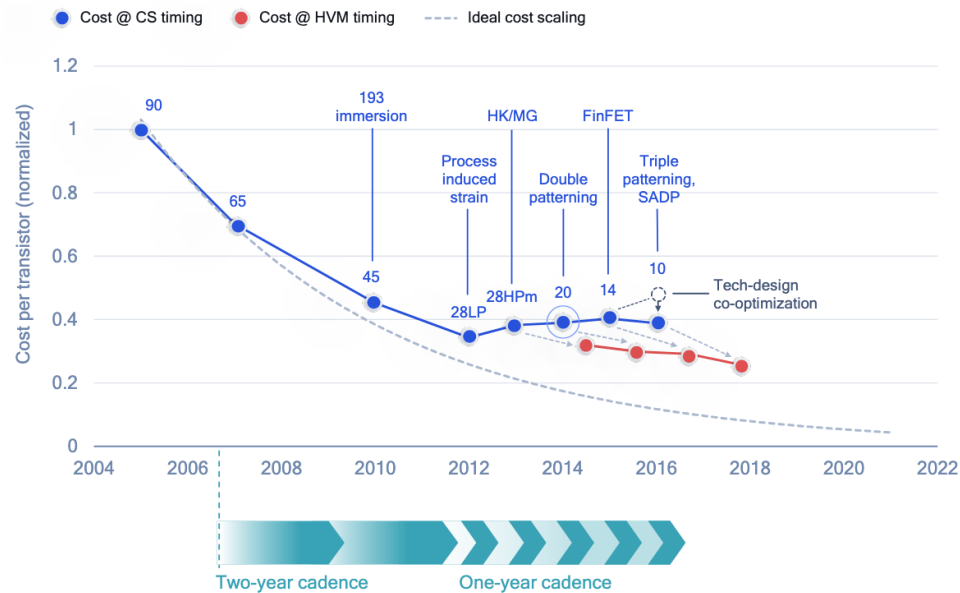
50 Years of Microprocessor Trend Data



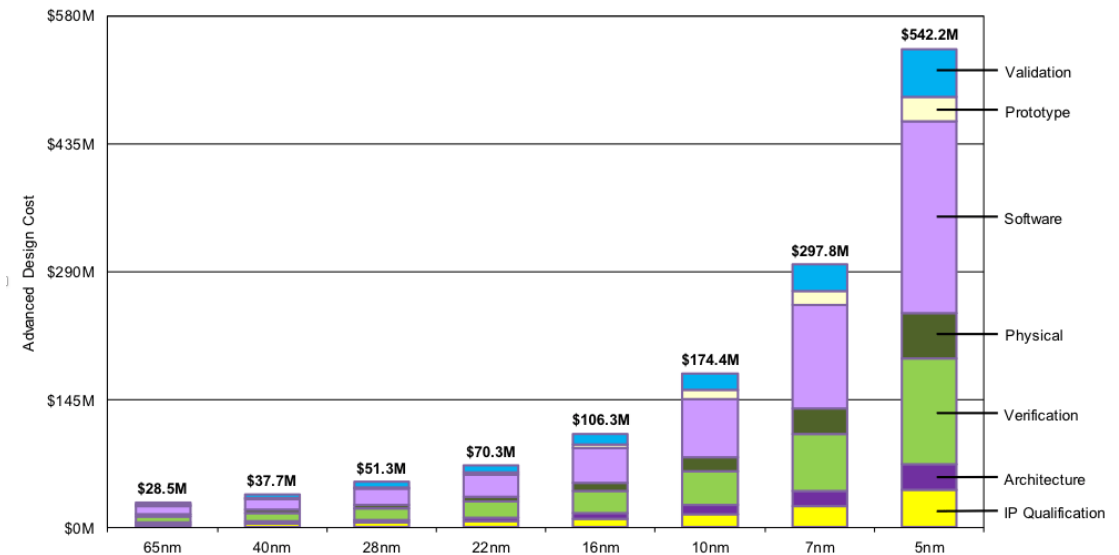
Original data up to the year 2010 collected and plotted by M. Horowitz, F. Labonte, O. Shacham, K. Olukotun, L. Hammond, and C. Batten
New plot and data collected for 2010-2021 by K. Rupp

Scaling made computation cheaper – until now

Cost per transistor trend

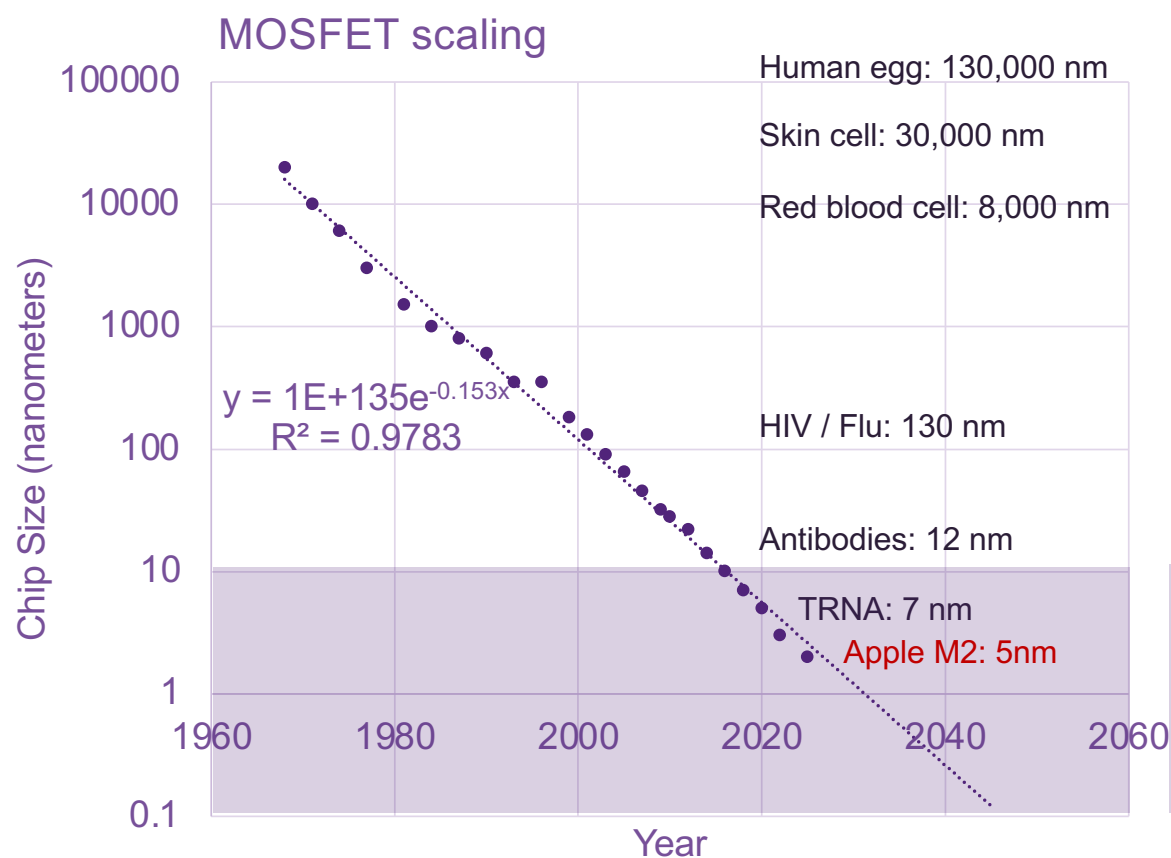


Source: Qualcomm
(https://erisummit.darpa.mil/docs/Mollenkopf_Steve_Qualcomm_Final.pdf)



Source: IBS
(<https://semiengineering.com/big-trouble-at-3nm/>)

Chip size of less than 10nm has quantum effects



“big” ~ more classical

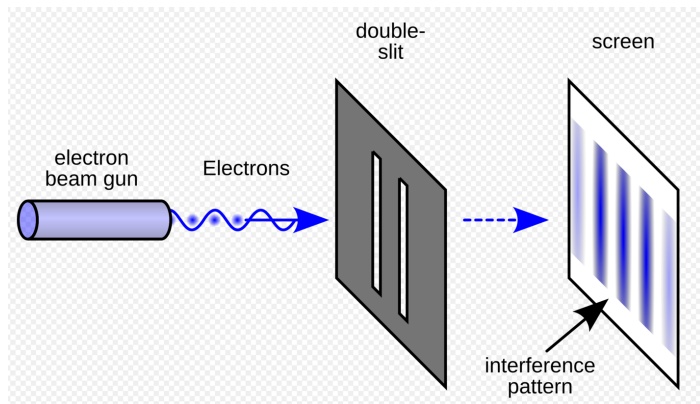


“small” ~ more quantum

Quantum tunneling leads to leakage current
and energy loss < 10nm

How will we make a computer from quantum rules?

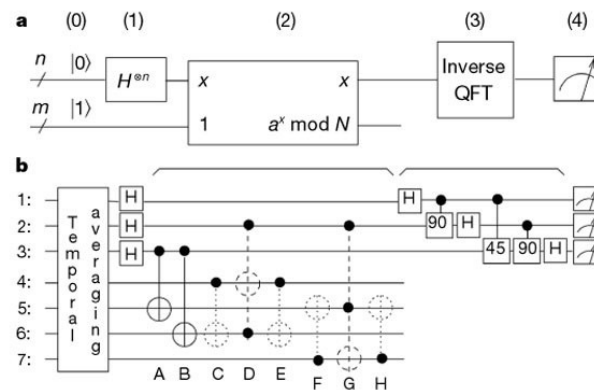
1920s: Quantum mechanics
confirmed in experiments



https://en.wikipedia.org/wiki/Double-slit_experiment#/media/File:Double-slit.svg

We see strange things in nature when
things become very small

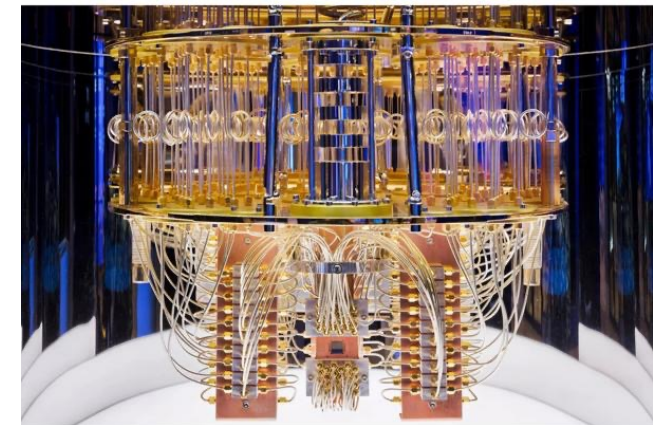
1970s – 1990s
Quantum algorithms proposed



<https://www.nature.com/articles/414883a>

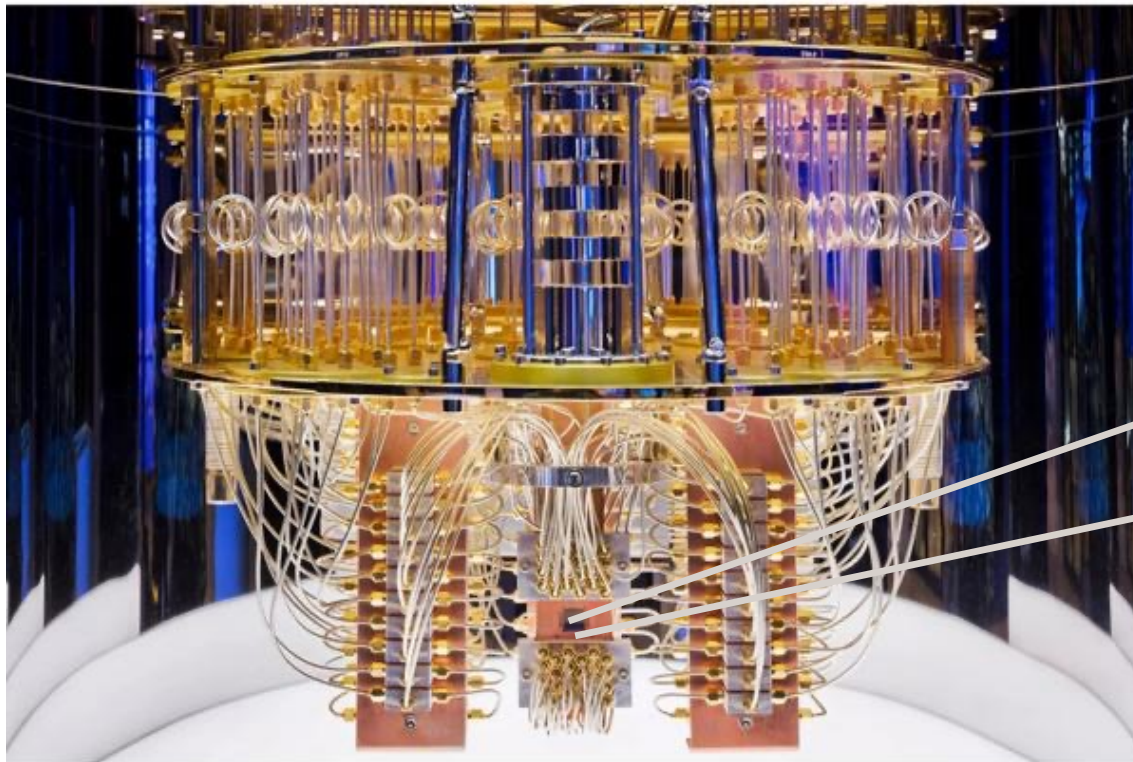
We describe quantum computing via
mathematics

2010 - 2016
Quantum computers

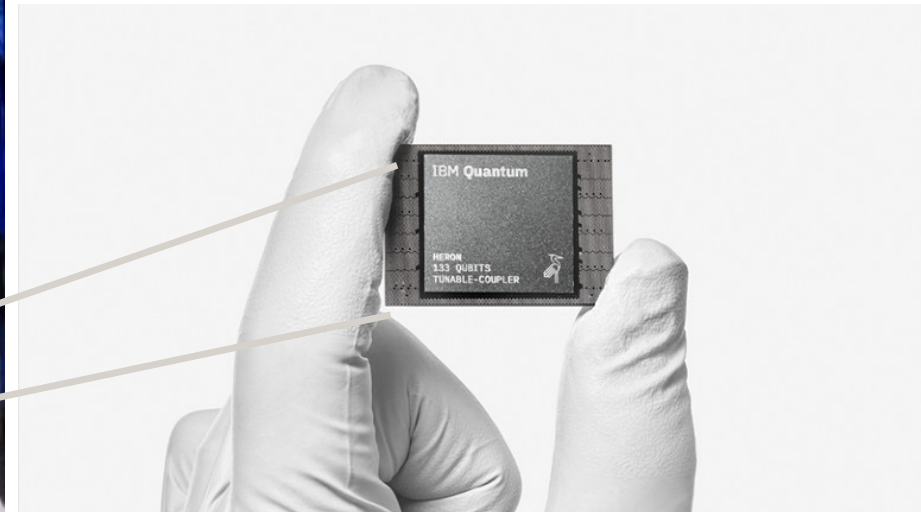


We build the hardware

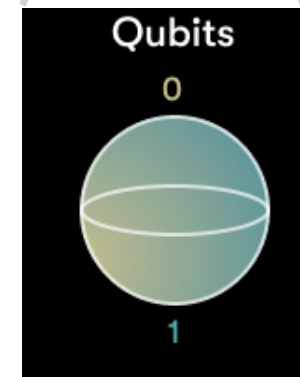
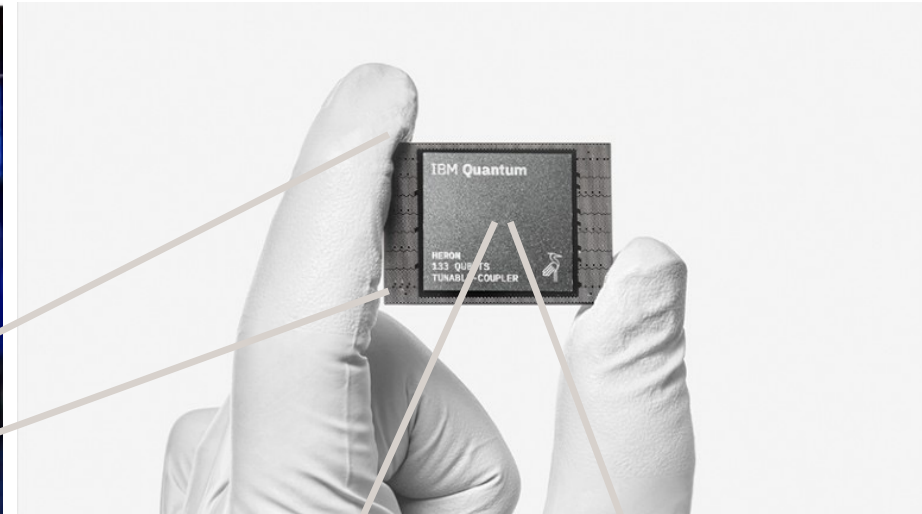
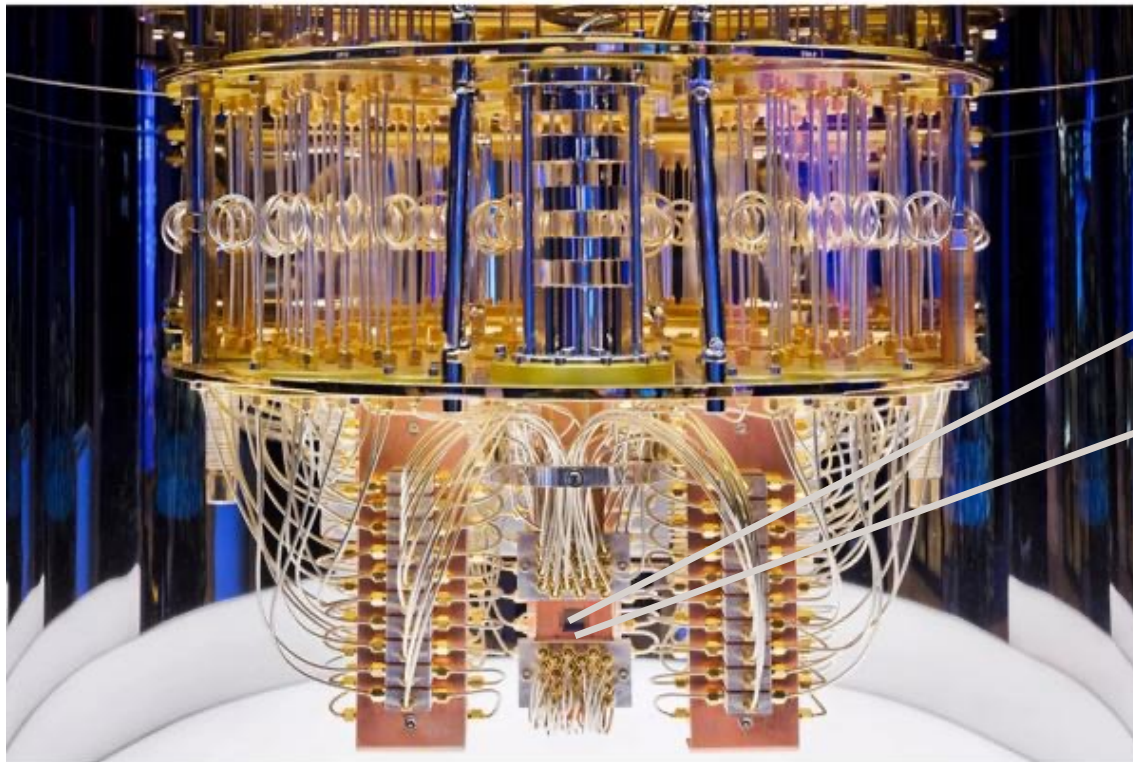
Today's quantum computer



Very, very small
Very, very cold

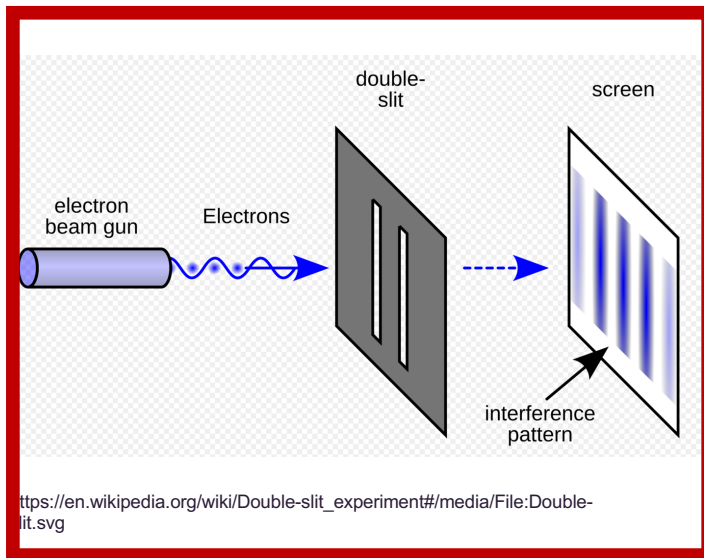


Today's quantum computer



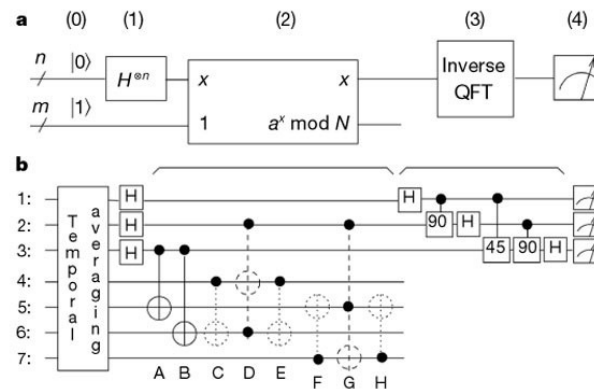
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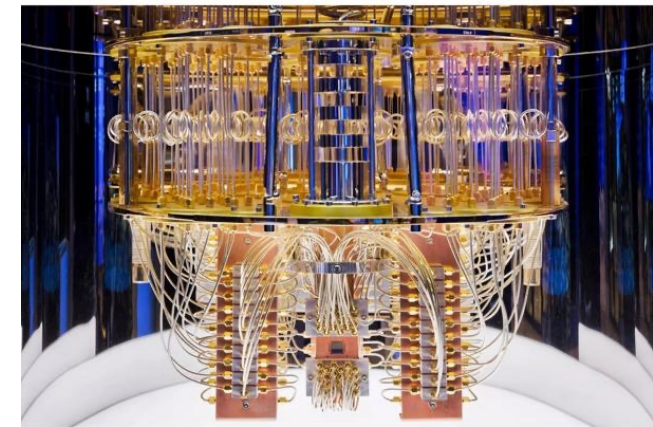
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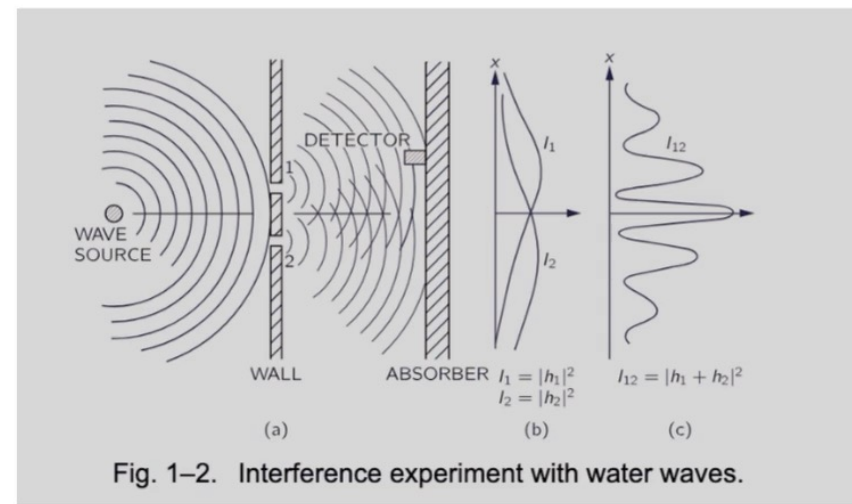
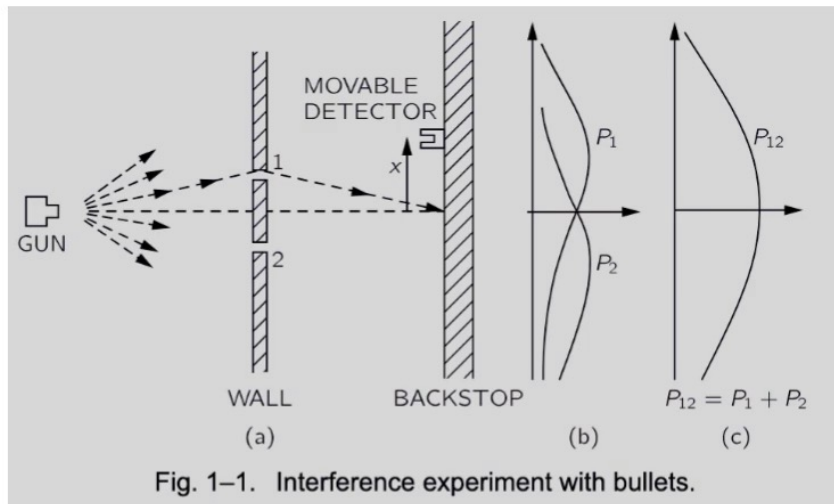
We describe quantum computing via
mathematics

2010 - 2016
Quantum computers



We build the hardware

Classical Physics



Original video: Quantum Made Simple (<https://toutestquantique.fr/en/>)

Edits to video: Dr. Riddhi S. Gupta (for educational correctness)

Added Figures: Feynman Lectures Vol 3 Ch 1

Quantum postulates

1. All knowledge of a **system** is contained in an abstract quantum state, $|\Psi\rangle$.
2. We cannot directly observe this $|\Psi\rangle$, or how $|\Psi\rangle$ evolves in time i.e. the *time evolution* of a system is distinct from *observing* a system in quantum physics
3. Observations are inherently random and perfect measurements collapse $|\Psi\rangle$ to take a known value
4. Uncertainty ‘protects’ quantum mechanical behavior of a system.

Quantum mechanics give us a manner to *allocate probabilities* to *observations* using information in $|\Psi\rangle$.